

Total No. of Questions : 8]

SEAT No. :

PB2368

[Total No. of Pages : 2

[6263]-218

B.E. (Mechanical Engineering)

PRODUCT DESIGN & DEVELOPMENT

(2019 Pattern) (Semester-VII) (Elective-IV) (402045A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Use of electronic pocket calculator is allowed.*
- 4) *Assume suitable data if necessary.*
- 5) *Figures to the right indicate full Marks*

- Q1)** a) Explain subtract and operate procedure for product teardown process. [7]
b) Describe in detail benchmarking. [6]
c) Explain Product characteristics in product analysis [4]

OR

- Q2)** a) What is concept selection? Explain Pugh's chart with example. [7]
b) Write a short note on SWOT analysis for a selection of profitable product. [6]
c) What is product policy of an organization? List down various product policies. [4]

- Q3)** a) What are the basic principles of dimensioning [7]
b) Explain BOM with example. [6]
c) Define Limit, Tolerance and Fit [4]

OR

- Q4)** a) What is product architecture? Explain types of product architecture. [7]
b) List down and explain different elements of production drawing. [6]
c) Define Tolerance? describe the types of Tolerances. [4]

P.T.O.

- Q5)** a) List down different design simulation techniques and explain any one. [8]
- b) What is Rapid prototyping? Define and enlist various methods of prototyping. [6]
- c) Explain Simulation driven design. [4]

OR

- Q6)** a) Write on make vs buy decision in product development with example. [8]
- b) Write note on Additive manufacturing [6]
- c) Differentiate Product testing and product validation. [4]
- Q7)** a) Write a short note on Design for robustness. [8]
- b) Write a short note on Product data Management (PDM). [6]
- c) Discuss the elements of PLM in detail. [4]

OR

- Q8)** a) Explain FMEA with example. [8]
- b) Differentiate Value analysis and value engineering. [6]
- c) Write a short note on Product lifecycle management (PLM). [4]

